

The Mechanistic Fixed-Effects Topic Model: Recovering Corporate Culture from Employee Reviews

Dan Biderman¹, David Blei², Wei Cai², Andrea Ciccarone², Amir Feder³, and Andrea Prat²

¹Stanford University

²Columbia University

³The Hebrew University of Jerusalem

This version: September 16, 2026

Latest Version

Abstract

Probabilistic topic models are widely used to uncover latent themes in observational text, but comparisons across groups are vulnerable to environmental variation in language. When documents are generated in different contexts, such as across industries or sentiment polarities, standard topic models conflate contextual differences with topic content, while bag-of-words representations discard contextual nuance. We introduce the mechanistic Fixed-Effects Topic Model (mFETM), an unsupervised generative model that explicitly separates invariant topic content from environment-specific deviations. To retain context and tone while preserving interpretability, mFETM defines topics as distributions over interpretable semantic concepts extracted from a large language model using sparse autoencoders. Applied to over 900,000 pro and con documents from Glassdoor reviews of S&P 1500 firms across 20 industries, mFETM improves held-out document perplexity while yielding topics that are substantially less confounded by review sentiment and industry vocabulary. In downstream firm-level analyses, mFETM recovers stable dimensions of corporate culture that reliably predict inclusion in Fortune’s 100 Best Companies to Work For. Finally, because the model operates in representation space, steering along estimated fixed-effect directions enables controlled counterfactual generation of workplace text across environments.

1 Introduction

Social scientists use topic models to turn large text corpora into interpretable measures of latent constructs (Grimmer and Stewart, 2013; Gentzkow et al., 2019). Applications span political agendas, legal briefs, managerial behavior, and central-bank deliberations (Quinn et al., 2010; Sim et al., 2015; Hansen et al., 2018; Bandiera et al., 2020). Topic models can recover ideologies, narratives, and organizational practices that are otherwise difficult to observe at scale.

Table 1: Two Reviews Describing Rigid Work Schedules

Company	Glassdoor Review
Amazon	<i>“Ran like the military, very cutthroat environment. You are given rates you need to pick and pack that are very hard to make if you don’t have the work in front of you. Thirty minute lunches and 15 minute breaks however you must walk about 5–10 minutes to get your lunch (if it’s not stolen) so even that you don’t look forward to.”</i>
Aflac	<i>“Being on the phone all day is kind of like jail. Pre-scheduled breaks and lunch times. An adult should be able to choose when they eat lunch. Sometimes they scheduled your lunch after you’d only been at your shift for an hour or two. Then you are stuck at your desk the rest of the day. Getting time off can be a hassle. It’s determined by a computer, not your supervisor. Very little to no advancement opportunities.”</i>

Corporate culture is a natural application. Persistent differences in how organizations operate may help explain performance gaps even after observable inputs are held fixed (Schein, 1985; Syverson, 2004; Kotter and Heskett, 2011; Gibbons and Henderson, 2013). Although a long literature classifies these differences into recurring cultural types (Handy, 1978; Cameron and Quinn, 2011), culture remains difficult to measure at scale. Employee-review websites are particularly useful in this context, as they provide a direct record of how workers experience their firms (see, e.g., Cai et al., 2024). Turning those reviews into comparable measures of corporate culture, however, raises two problems for traditional topic models.

The first is environmental heterogeneity. In a corpus drawn from many sources, a common theme may be expressed differently across industries, firms, or document types. A worker describing rigid schedules may mention pick rates and timed breaks in a warehouse, but call queues and computer-assigned time off in an office. A standard topic model can interpret these differences in expression as differences in substance. We refer to this source-specific language as an environmental language fixed effect.

The second problem is representation. Traditional topic models (Blei et al., 2003) treat each document as a bag of word counts. Under this representation, structure is captured by word co-occurrence, so potentially important semantic objects, like context and tone, are discarded. Corporate concepts often appear through combinations of phrases, metaphors, and evaluations rather than through a single keyword, which limits a bag-of-words representation for this task. Word-based topics are also difficult to interpret, as their meaning must be inferred from a ranked list of words.

The two reviews in Table 1 make both problems concrete. A warehouse associate at Amazon and a customer-service specialist at Aflac both complain about rigid schedules and tightly regulated routines. Yet the first describes rates, packing, and breaks, while the second describes phones, desks, and computer-assigned time off. Their common complaint is expressed through language,

metaphor, and tone that is industry or task specific, and that simple word counts can only partially capture.

We develop a topic model that addresses both problems jointly, the mechanistic Fixed-Effects Topic Model (mFETM). The model has two components. First, mFETM places fixed effects inside the generative process: a global backbone captures what a topic shares across reviews, while environment-specific deviations capture how it is expressed in a particular industry, review side, or other observed environment. Second, it replaces the bag-of-words representation with a bag of concepts. This representation is built from counts of automatically labeled semantic features extracted from a language model by a sparse autoencoder (SAE). A topic is therefore a distribution over readable concepts rather than words.

To predict language, a large language model develops internal representations of recurring meanings, such as schedule flexibility, supportive coworkers, or customer service. These meanings are mixed together across many coordinates and are therefore difficult to read directly. An SAE decomposes those dense internal representations into a sparse set of recurring features, meaning that for any passage only a small number are active. Passages that strongly activate the same feature reveal its common meaning, allowing the feature to be labeled and used as an interpretable concept.

We estimate mFETM on a large corpus of Glassdoor reviews and compare it with a mechanistic vanilla topic model, mVTM. The benchmark uses the same concept representation and the same number of topics but omits the fixed effects.¹ The comparison holds the data, representation, topic count, and encoder architecture fixed while assessing the contribution of incorporating observed environments during topic estimation. We call the model mechanistic because its estimated topics and environment shifts live in the language model’s representation space and can be manipulated directly during generation.

We report three sets of results. First, we test the property the fixed effects are designed to deliver: topics should vary less across observed environments and reveal less about them. Between positive and negative reviews, mFETM’s average topic-share spread is less than half mVTM’s. The spread is also lower across industries, although to a smaller degree. In direct prediction exercises, mVTM’s topics explain 55.4% of review polarity, compared with 19.7% for mFETM, and predict industry better in all 28 pairwise sector comparisons (Section 5). Comparisons across firms therefore inherit less environmental variation from mFETM topics.

Second, we use the topics to predict inclusion on Fortune’s 100 Best Companies to Work For list, an external benchmark of employer quality. Normalization-invariant topic contrasts show that mFETM provides a broader, more interpretable range of associations spanning compensation, work-life balance, coworker relations, management, restructuring, and workplace amenities. mVTM also identifies general workplace attributes, but mixes them with environment-specific predictors: technology strategy, for example, is strongly associated with industry. It therefore provides a less complete map of the general workplace dimensions that distinguish top employers. A sharper

¹Zheng et al. (2025) establish the advantage of the concept representation over bag-of-words for vanilla topic models; that is not strictly the comparison studied here.

comparison asks where each model’s overall fit originates. We sequentially remove the topics most associated with industry and review polarity and re-estimate the Fortune model. Once 10% to 30% of topics are removed, mFETM retains substantially more of its original fit than mVTM. Its ability to distinguish top employers therefore rests more on general workplace themes and less on sector- or tone-specific topics.

Finally, combining the SAE representation with the fixed-effects model makes topics and environment shifts directions in the language model’s representation space, and we exploit this structure by steering generation along those directions. Given a topic, we steer the model along an estimated environment direction and examine how the same managerial theme is expressed in a different industry, or as a positive rather than a negative review, with the theme held fixed. Because the direction is estimated from the data, the fixed effects become objects we can manipulate as well as statistical controls.

We return to the Amazon and Aflac reviews to show how the modeling choices work. Table A1 reports historical bag-of-words estimates that motivated the project; they are not the current $K = 50$ comparison. In that illustration, a vanilla topic model splits the reviews along industry and polarity lines, despite their common complaint. Adding fixed effects places both on the same leading topic by absorbing some industry and polarity language. Yet the shared topic remains a word list (workplace, lunch, food) from which the underlying complaint must still be inferred.

The bag of concepts addresses this second limitation. Building on Zheng et al. (2025), we represent each document by the labeled features it activates inside a language model (Section 2). Topics become distributions over those concepts. Table A2 shows the gain in readability by pairing the leading word- and concept-based topics by rank for exposition; the rows are not aligned latent objects. The concept lists make the semantic content used by the model explicit.

After estimation, we give each concept list a short name. A language model removes up to two outlying descriptions, combines the remaining concepts with 20 reviews that have the highest topic shares, and names the narrowest workplace construct supported by both sources. The name is only a label; it never enters estimation (Section 4.1 and Appendix D). The current $K = 50$ estimates do not place the Amazon and Aflac reviews on the same leading topic: Amazon loads most heavily on physical warehouse operations and Aflac on schedule inflexibility, although two topics appear in both top-three sets (Table 2). The pair motivates the measurement problem, and the aggregate comparisons test the model.

Together, these steps form a measurement pipeline for corporate culture. Fixed effects separate shared themes from environmental language; the SAE represents reviews as mixtures of interpretable concepts; and the final naming step turns related groups of concepts into readable workplace constructs. By separating what workers discuss from the language of their industry or review side, mFETM reveals dimensions of corporate culture that a conventional topic model leaves bundled with context and sentiment.

The paper connects three literatures. First, work measuring corporate culture from text and relating it to firm outcomes (e.g., Guiso et al., 2015; Li et al., 2021; Cai et al., 2024); we contribute

Table 2: Top mFETM Topics for the Two Reviews

Topic (mFETM)	Share
Amazon	
Warehouse Operations and Equipment Handling	0.257
Attendance and time-off policies	0.124
Wages and Working Hours	0.116
Aflac	
Attendance and time-off policies	0.278
Warehouse Operations and Equipment Handling	0.172
Variable work schedules and hours	0.109

Notes. Each review’s top three mFETM topics from the estimated model. Share is the review’s estimated topic share. Two topics appear in both top-three sets, but the leading topic differs.

an unsupervised and interpretable measure designed to reduce contamination by industry and review polarity. Second, topic models that place document covariates in the generative process, including structural topic models, sparse additive models, and neural variants (Roberts et al., 2014; Eisenstein et al., 2011; Card et al., 2018); mFETM estimates these effects over concepts rather than words. Third, work showing that language-model activations encode concepts as directions and that sparse autoencoders can recover them (Bricken et al., 2023; Zheng et al., 2025); we put these tools to a new use by turning estimated, topic-specific environment shifts into manipulable directions for controlled generation.

The rest of the paper proceeds as follows. Section 2 describes the Glassdoor data and construction of the concept representation. Section 3 presents the model, inference, and mVTM benchmark. Section 4 illustrates the estimated topics and compares held-out fit. Section 5 tests how strongly the topics remain associated with the observed environments. Section 6 predicts Fortune-list inclusion and traces each model’s fit to its source. Section 7 steers generation along the estimated fixed-effect directions, and Section 8 concludes.

2 Representing Text with Bags of Concepts

We represent each document by counts of labeled semantic concepts rather than counts of words. This preserves information that word counts miss and makes the estimated topics more directly readable. Building on Zheng et al. (2025), both the concept vocabulary and the document-level counts come from a pretrained sparse autoencoder.

2.1 Glassdoor Data

We analyze employee reviews from Glassdoor, an online platform established in 2008 on which current and former employees anonymously review their workplaces. The underlying dataset is a firm-year panel of reviews for S&P 1500 firms used in prior research (Cai et al., 2024; Lee et al., 2021; Green et al., 2019). The source extract spans 2008–2022, contains millions of original

reviews, and covers more than 190,000 job titles, with rank-and-file employees far outnumbering senior executives (Karabarbounis and Pinto, 2018). Two platform features may mitigate selection and authenticity concerns: a “give-to-get” policy broadens participation (Marinescu et al., 2021), and accounts are linked to real names even though reviews are published anonymously (Green et al., 2019).

The reviews provide rich descriptions of workplace experience across firms and industries. They are therefore well suited to testing whether mFETM can separate common cultural themes from the context-specific language used to describe them.

2.2 Sparse Autoencoders

Sparse autoencoders (SAEs) were developed to interpret large language models (Cunningham et al., 2023; Bricken et al., 2023). As a language model reads a review, it represents each token and its context with a high-dimensional activation vector a . Following the mechanistic-interpretability literature, we treat recurring semantic features as directions in that activation space (Bricken et al., 2023; Zheng et al., 2025). Many signals share the same coordinates, so individual coordinates are not directly interpretable (Elhage et al., 2022).

The SAE approximates a with a sparse, nonnegative combination of directions drawn from a larger dictionary of h candidates:

$$r = \sigma(W_{\text{enc}} a + b_{\text{enc}}) \in \mathbb{R}_{\geq 0}^h, \quad \hat{a} = W_{\text{dec}}^{\top} r + b_{\text{dec}}.$$

The first equation maps the residual-stream activation a into sparse feature coefficients r , one per dictionary direction. The activation function σ sets coefficients below a learned feature-specific threshold to zero, producing a sparse, nonnegative representation; this thresholding rule is called JumpReLU. The second equation reconstructs a as $\hat{a}$ by combining the rows of W_{dec} with weights r . The parameters are estimated to reconstruct a accurately while keeping most entries of r at zero.

In practice, this fit-sparsity trade-off often produces directions aligned with individual concepts, such as mentions of long commutes or expressions of gratitude (Bricken et al., 2023). At any point in the text only a small number of entries of r are positive; when one is positive, we say the corresponding concept fires.

The coordinates are interpreted and validated empirically. A language model reads passages where a coordinate fires strongly and summarizes their common content in a short label. The label can then be evaluated by how well it predicts activations on new text (Bills et al., 2023). We call each labeled coordinate a concept. Table 3 lists the concepts that fire in the largest share of documents in our corpus.

We use a pretrained SAE and hold it fixed, so its concepts reflect general language rather than our corpus. Each concept comes with a previously generated label. Passing a review through the

Table 3: Most Common Concepts in the Corpus

Concept	Share of documents
Elements related to personal connections and experiences in social contexts	74.0%
Elements related to shopping experiences and customer service	62.9%
Mentions of job performance and employment situations	54.7%
Aspects related to food quality and dining experiences, particularly emphasizing price, taste, and service	51.9%
References to family relationships and genealogical connections	48.9%
Various positive adjectives and expressions of approval	46.7%
Issues related to ethics and accountability in business practices	43.7%
Terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA)	42.8%
Criticisms of food quality and service in restaurant reviews	41.1%
Positive descriptors related to quality and performance	40.7%

Notes. The ten concepts firing in the largest share of documents in the final thresholded count matrix, with their labels, after the pruning of Section 2.3.

language model and SAE records which concepts fire at each token; their counts over the document form its bag of concepts.

2.3 Constructing the Bag of Concepts

We next convert the raw reviews into the bag-of-concepts count matrix used by the topic models. We summarize the procedure here and report implementation details in Appendix C.

We treat each review’s “pros” and “cons” sections as separate documents carrying a polarity label. After the document-level filters, the corpus contains 906,938 documents: 453,215 pros and 453,723 cons.²

Each document is read by Gemma-2-9B, an open language model (Gemma Team, 2024). At each token, we take the internal representation a described in Section 2.2 and pass it through a pretrained Gemma Scope SAE (Lieberum et al., 2024). This yields the concepts that fire and their strengths. We prune concepts that are too rare or common to be informative, along with those describing syntax rather than content, and count how often each surviving concept fires strongly in each document:

$$c_{d,v} = \sum_j \mathbf{1}\{\alpha_v(a_{d,j}) > q_v\},$$

where the sum runs over the tokens of document d , $\alpha_v(a_{d,j})$ is concept v ’s firing strength at token j , and q_v is a concept-specific threshold. Appendix C reports the SAE layer, pruning rules, and thresholds. The final vocabulary contains $V_{\text{kept}} = 4,735$ labeled concepts; the counts $c_{d,v}$ are the inputs to both topic models.

²The reviews are a 20% random sample of the corpus described in Section 2.1, restricted to the 20 two-digit NAICS industries with more than 1,000 documents.

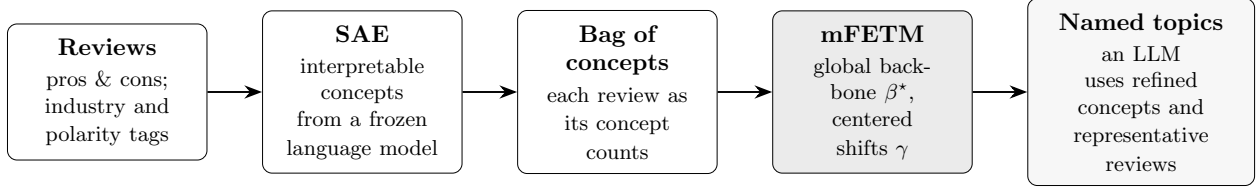


Figure 1: The pipeline. A sparse autoencoder maps each review to the interpretable concepts it expresses; mFETM estimates a shared-topic backbone β^* , identified relative to centered environment deviations γ , on the resulting concept counts. Each topic is then named from its refined concepts and representative high-share reviews. mVTM is the same model with $\gamma \equiv 0$.

3 The Mechanistic Fixed-Effects Topic Model

The previous section turned each review into counts of interpretable concepts. Pooled topic estimation can still absorb environment-specific language into the resulting measures. We now formalize that problem and specify a model that addresses it during estimation. Figure 1 summarizes the full pipeline.

3.1 The Fixed-Effects Problem

Social scientists often use topic shares as downstream measures, comparing what is discussed across firms, industries, or time. Such comparisons require a common meaning across environments, which pooled topic models need not preserve.

Consider a workplace-conditions theme. Industry vocabulary and review polarity may split it into sector-specific variants or fuse it with complaint language. The resulting shares are not directly comparable: differences across sectors may reflect vocabulary rather than substance. Their measurement error is systematic because it depends on how each environment splits and relabels the themes. Nor do the shares necessarily index a common latent construct, making counterfactual comparisons across environments ill-defined. Downstream controls cannot repair a distortion embedded in the document-level measure itself.

mFETM addresses the problem where it arises. Fixed effects absorb systematic contextual language during estimation, leaving a shared topic component, more comparable topic shares, and deviations that describe how each environment expresses a theme.

3.2 Model Specification

The input of the model we build is the corpus of Section 2, $\mathcal{D} = \{(\mathbf{c}_1, \mathbf{x}_1), \dots, (\mathbf{c}_n, \mathbf{x}_n)\}$: each document carries its concept counts $\mathbf{c}_i \in \mathbb{Z}_{\geq 0}^{V_{\text{kept}}}$ and a fixed-effects vector $\mathbf{x}_i \in \{0, 1\}^E$ marking the environments it belongs to. In our main specification these are the firm’s two-digit NAICS industry and the review’s polarity, so $E = 22$: 20 industries and 2 polarities. In the model, a concept token is one recorded feature-firing event; one text token can generate more than one such event.

Each document is a mixture of K topics. The vector θ_i contains document-specific topic logits, and $\pi_i \equiv \text{softmax}(\theta_i)$ denotes the topic shares. Topic k has a global vector of concept logits, $\beta_k^* \in \mathbb{R}^{V_{\text{kept}}}$. For each environment e , $\gamma_{e,k} \in \mathbb{R}^{V_{\text{kept}}}$ shifts those logits. In our application every document activates one industry effect and one polarity effect. The generative process is

1. draw global topic logits $\beta_k^* \sim \mathcal{N}(0, I)$ and auxiliary environment coefficients $u_{e,k}$ from the shrinkage prior below, then set the centered deviations $\gamma^g = P_g u^g$, for $k = 1, \dots, K$;
2. for each document i with environment $\mathbf{x}_i$, draw topic logits $\theta_i \sim \mathcal{N}(0, I)$, and for each concept token j :
 - (a) draw a topic assignment $z_{ij} \sim \text{Cat}(\pi_i)$;
 - (b) draw a concept $c_{ij} \sim \text{Cat}(\text{softmax}(\beta_{z_{ij}}^* + \gamma_{z_{ij}} \cdot \mathbf{x}_i))$.

Here $\gamma_{z_{ij}} \cdot \mathbf{x}_i = \sum_e x_{i,e} \gamma_{e,z_{ij}}$ adds the shifts of the environments the document belongs to. Marginalizing the topic assignment z_{ij} , every concept token in document i is drawn from the same document-level mixture over concepts,

$$\eta_i = \sum_{k=1}^K \pi_{i,k} \cdot \text{softmax}(\beta_k^* + \gamma_k \cdot \mathbf{x}_i), \quad (1)$$

so the document's bag of concepts $\mathbf{c}_i$ is multinomial in η_i , with log-likelihood $\sum_v c_{i,v} \log \eta_{i,v}$ up to the document-specific multinomial coefficient. The shares π_i weight the topics, and within each topic the active environments shift the concept distribution.

The likelihood depends only on $\beta_k^* + \gamma_k \cdot \mathbf{x}_i$. Without a normalization, a component common to all industries or both polarities could be assigned either to the global topic or to the environment shifts without changing any fitted probability. We therefore divide γ into industry and polarity blocks, $g \in \{I, P\}$, and impose

$$\sum_{e \in g} w_e^g \gamma_{e,k,v} = 0 \quad \text{for every } (g, k, v).$$

Here w_e^g is environment e 's frequency within block g in the fit sample. We impose these constraints during estimation, so β_k^* is the global backbone relative to the centered environment deviations, up to the usual additive normalization of multinomial logits, and each $\gamma_{e,k}$ is an environment-specific deviation. The normalization does not restrict fitted probabilities: any unconstrained decomposition can be recentered into this form without changing $\beta_k^* + \gamma_k \cdot \mathbf{x}_i$. mVTM is the special case $\gamma \equiv 0$, in which the topic–concept distributions must absorb any environment-related language. Figure 2 shows the graphical model.

Centering identifies the decomposition; shrinkage separately controls the size of the deviations. We use a Gamma–Gaussian shrinkage construction. For every environment–topic–concept coef-

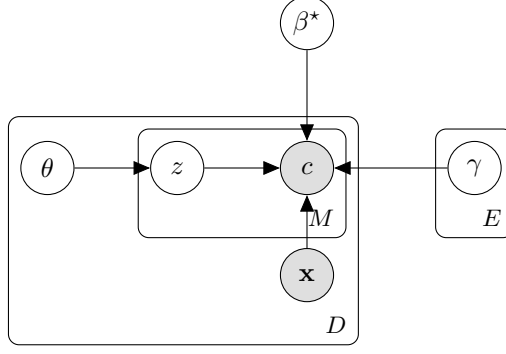


Figure 2: Graphical model for the mechanistic fixed-effects topic model. M indexes concept tokens, D documents, and E environment levels. c is an observed concept, z its topic assignment, θ the document’s topic-share logits, $\mathbf{x}$ its observed industry and polarity indicators, β^* the global topic backbone, and γ the centered environment deviations. Setting $\gamma \equiv 0$ recovers mVTM.

ficient, we draw a precision and a conditionally Gaussian auxiliary coefficient, then project each effect block onto the weighted-centered subspace,

$$\tau_{e,k,v} \sim \text{Gamma}(a_\tau, b_\tau), \quad u_{e,k,v} \mid \tau_{e,k,v} \sim \mathcal{N}(0, \tau_{e,k,v}^{-1}),$$

$$\gamma_{k,v}^g = [I - \mathbf{1}(w^g)^\top] u_{k,v}^g, \quad g \in \{I, P\}.$$

We set the shape parameter to $a_\tau = \exp(1.25)$ and the rate parameter to $b_\tau = \exp(0.50)$, implying mean precision $a_\tau/b_\tau = 2.117$. The projection then enforces the centering constraint by construction.³

3.3 Inference

Conditional on the realized precision field, we place the variational factor on the auxiliary coefficients u and obtain the centered deviations deterministically as $\gamma^g = P_g u^g$, where $P_g = I - \mathbf{1}(w^g)^\top$. The posterior over θ , β^* , and u is intractable, so we use amortized variational inference (Ranganath et al., 2014; Kingma and Welling, 2013; Blei et al., 2017) and integrate out z . A one-hidden-layer encoder maps each document’s concept counts to the mean and variance of a Gaussian approximation for θ_i ; the hidden layer has 50 units, ReLU activation, and batch normalization. The objective is

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{c} \mid \theta, \beta^*, \gamma(u))] - \sum_i \text{KL}\{q_\phi(\theta_i) \parallel p(\theta_i)\} \\ - \text{KL}\{q(\beta^*) \parallel p(\beta^*)\} - \text{KL}\{q(u) \parallel p(u \mid \tau)\}.$$
(2)

We estimate the variational mean and variance of β^* and the mean of u ; the fixed coefficient-level precision field sets the variance of u before centering. Each likelihood draw is mapped to $\gamma^g = P_g u^g$,

³We treat a_τ and b_τ as fixed tuning parameters rather than estimating them from the observed corpus, as an empirical Bayes procedure would. We draw the coefficient-specific precisions once and hold them fixed throughout estimation and evaluation.

and the mean of u is projected to weighted mean zero after every Adam update (Kingma and Ba, 2014). The document likelihood and θ KL are averaged within each minibatch; the global β^* and u KL terms are divided by the number of fit documents.

The comparison holds the data, representation, topic count, and encoder architecture fixed. Optimization schedules are selected separately by model, but both use the same validation-based checkpoint rule and are evaluated on the test split without further tuning. Appendix E gives the complete training and evaluation protocol.

3.4 Baseline and Comparisons

mVTM is the benchmark throughout: it uses the same representation, encoder architecture, and $K = 50$ topics, but no γ . The comparison is designed to assess what changes when observed environments enter the topic–concept distributions. We compare held-out fit, test how strongly each model’s topic shares remain associated with the environments, and ask whether they retain firm-level information.

4 Topics and Predictive Fit

We estimate mFETM and mVTM with $K = 50$ topics on the same concept representation. The fixed effects first change what a topic represents. Instead of forcing common content and environment-specific language into one topic distribution, mFETM can recover a shared theme and show separately how it is expressed in different settings. We first make that decomposition concrete with a matched topic, then verify that it does not cost predictive fit. The dispersion diagnostics use the test split; the firm-year analyses aggregate the full corpus.

4.1 An Estimated Topic

To show what the models return, we examine the mFETM topic named “Enterprise Project Management and Integration.” The full lists of topic names are in Appendix B. The example contrasts the matched mVTM topic with mFETM’s global component and two fixed-effect refinements.

Each concept already has an automatic label. To name topic k , GPT-5.6 Luna receives 10 concepts selected from the 12 largest global weights and 20 short reviews with the highest topic shares evaluated at the posterior-mean logits. The excerpts anchor general-web concept labels in their workplace context. Environment labels and refinements do not enter the name, and naming occurs entirely after estimation. Appendix D gives the exact selection rule and prompt.

The design holds a single semantic theme fixed and contrasts how the models represent it. Table 4 reports four versions: the mVTM topic; mFETM’s global backbone β^* ; the Con refinement $\beta^* + \gamma_{\text{Con}}$; and the Information-sector refinement $\beta^* + \gamma_{\text{NAICS}_{51}}$. Each refinement adds one centered deviation before the concept distribution is formed. The mVTM counterpart, “Enterprise Technology Strategy and Innovation,” is matched on the cosine between the two row-centered concept-logit vectors over the shared 4,735 concepts. Its cosine with the mFETM topic is 0.79; the next two

Table 4: One estimated topic, mFETM’s “Enterprise Project Management and Integration,” seen four ways: its mVTM counterpart “Enterprise Technology Strategy and Innovation” (matched by cosine of the row-centered concept-logit vectors), the global mFETM backbone β^* , and its Con and NAICS 51 (Information) refinements, each listed by its four top SAE concept-features.

	Top concepts
mVTM β	references to client experiences and services, terms related to information technology management, terms related to digital technology and information management, concepts related to multicultural strategies and organizational excellence
mFETM β^*	phrases related to professional roles and responsibilities, concepts related to multicultural strategies and organizational excellence, references to strategies and roles related to corporate sustainability and management, key terms and phrases related to employment and job responsibilities
mFETM $\beta^* + \gamma_{\text{Con}}$	phrases related to professional roles and responsibilities, terms related to information technology management, references to strategies and roles related to corporate sustainability and management, concepts related to multicultural strategies and organizational excellence
mFETM $\beta^* + \gamma_{\text{NAICS 51}}$	phrases related to professional roles and responsibilities, concepts related to multicultural strategies and organizational excellence, content related to business opportunities and strategies in sustainability and biotechnology, references to strategies and roles related to corporate sustainability and management

matches are 0.59 and 0.57. If the fixed effects work as designed, they should isolate what is common across environments from what is specific to polarity or industry, holding the underlying semantic object constant.

The mVTM topic combines client services, information-technology management, digital information systems, and organizational strategy in a single distribution. mFETM’s global backbone instead emphasizes professional roles, organizational excellence, corporate management, and job responsibilities. The refinement rows then isolate the environmental margin: the Con refinement brings information-technology management into the top four, while the NAICS 51 refinement brings sustainability and biotechnology strategy into the top four. Professional roles remains the leading concept in the global, Con, and NAICS 51 versions. The underlying managerial theme remains recognizable, while industry- and polarity-specific expressions are made explicit rather than allowed to redefine the topic itself. The global backbone can therefore serve as a more comparable measure across firms, and the refinements retain the contextual language as separately interpretable objects. Appendix F reports complementary geometric results; we next ask whether this decomposition sacrifices predictive fit.

4.2 Predictive Fit: Perplexity

The decomposition is useful only if separating common from environment-specific content does not make the model worse at representing documents. Both models use $K = 50$ topics on the same concept representation and the same document-level fit/validation/test partition; the fixed effects are the structural difference. We evaluate true document-completion perplexity on the held-out test split ($n = 90,348$). A Bernoulli half-split assigns each concept firing to the observed

set with probability 1/2; the encoder infers topic shares from those counts, and the remaining firings are scored. The same realized split is used for both models. mFETM reaches 680.5 against mVTM’s 694.6, a 2.0% reduction. The advantage appears in every reported slice (Figure 3): 589.2 against 609.7 on pros, 754.4 against 762.4 on cons, 564.0 against 603.5 in Accommodation and Food (NAICS 72), 717.3 against 727.6 in Information (NAICS 51), and 740.5 against 749.6 in Manufacturing (NAICS 33).

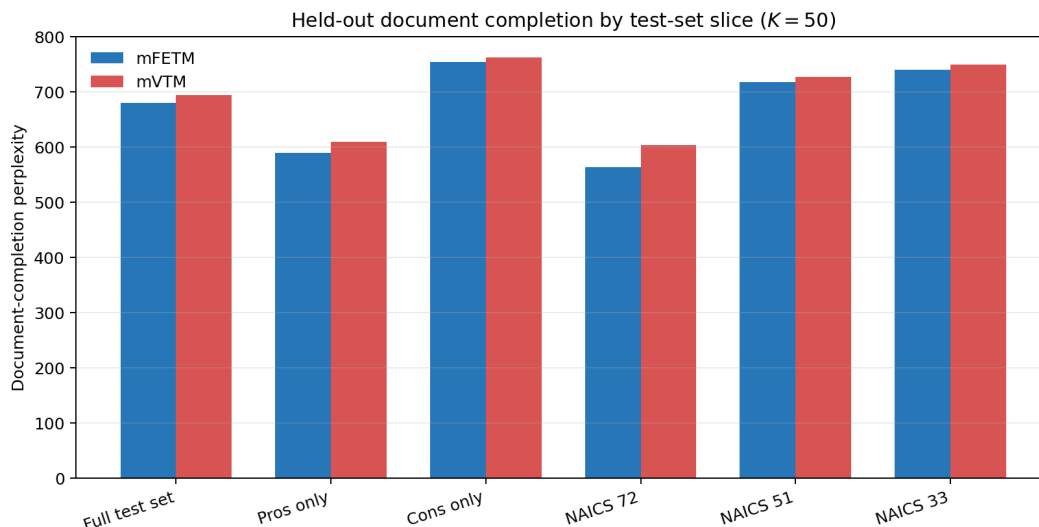


Figure 3: Held-out document-completion perplexity by test-set slice at $K = 50$ (lower is better). A Bernoulli half-split assigns each concept firing to the observed set with probability 1/2; both models score the same remaining firings.

At test time mFETM conditions on each document’s observed industry and polarity, while mVTM has no corresponding covariates. The 2.0% reduction is therefore the predictive gain from conditioning the topic–concept distributions on observed environments. mFETM also does not obtain cleaner global topics by discarding information. It fits held-out documents better overall and in each displayed environment. We next test whether the resulting global topics are in fact less tied to those environments.

5 How Environment-Associated Are the Topics?

If a topic intended to measure corporate culture also reveals whether a review is positive or negative, or which industry produced it, comparisons across firms inherit those environmental differences. mFETM is designed to move that information into the fixed-effect refinements and leave the global topic shares more portable. We test this implication in two ways: how much average topic shares move across environments, and how accurately the full topic vector predicts the environment itself.

5.1 Topic-Share Dispersion

We first ask whether mFETM topic shares vary less across observed environments. Pooled models may turn contextual terms such as “claims” in insurance or “fulfillment” in retail, as well as pro/con polarity, into topic content; mFETM assigns their systematic component to γ . Lower dispersion indicates more portable topics while allowing genuine theme prevalence to differ across industries and review sides.

For each topic k , we compute its average share $\bar{\pi}_{ke}$ in environment e and the spread $\Delta_k = \max_e \bar{\pi}_{ke} - \min_e \bar{\pi}_{ke}$. We calculate the measure across NAICS-2 industries with at least 1,000 documents and across pro/con on the test split. A small spread indicates that a topic is used at similar rates across environments; a large spread may reflect either contextual language or genuine differences in theme prevalence.

The measure is transparent and requires no topic alignment across models: we summarize each model’s topics separately and compare the two distributions of $K = 50$ spreads. Because the statistic uses the most and least prevalent environments, it is deliberately sensitive to extremes. We use a two-sided Welch two-sample t -test as a topic-level reference test, not as document-level sampling inference: topic shares within a model are jointly estimated and sum to one. Interquartile-range versions, which downweight extremes, also shift downward under mFETM.

Figure 4 reports the distribution of the spreads across topics for both models, for industries (left panel) and review side (right panel), and Table 5 the test. Across industries the average spread falls from 1.31 percentage points under mVTM to 1.03 under mFETM. The reduction points in the intended direction, although the topic-level reference test does not distinguish the two distributions at the 5% level ($t = -1.72$, two-sided $p = 0.090$).

The contrast is much sharper between positive and negative reviews. mVTM topics display an average spread of 1.18 percentage points between pros and cons. mFETM cuts this gap to 0.43, a reduction of nearly two thirds (topic-level Welch $t = -2.98$, two-sided $p = 0.004$). The model is therefore substantially less likely to treat the tone of a review as the topic being discussed. For corporate-culture measurement, dissatisfaction with management and management itself then become separable objects rather than one sentiment-charged topic.

5.2 Predicting the Environment from Topics

Dispersion considers one topic at a time. A stronger test asks how much of the environment can be reconstructed from the entire topic vector. If the fixed effects have separated context from content, even the topics taken jointly should be weaker predictors of industry and review side. We first regress the pro/con indicator on the topic shares over the full corpus ($n = 906,938$),

$$\text{pro}_i = \alpha^m + \sum_{k=1}^{K-1} \delta_k^m \hat{\pi}_{ik}^m + \varepsilon_i^m, \quad m \in \{\text{mFETM}, \text{mVTM}\}. \quad (3)$$

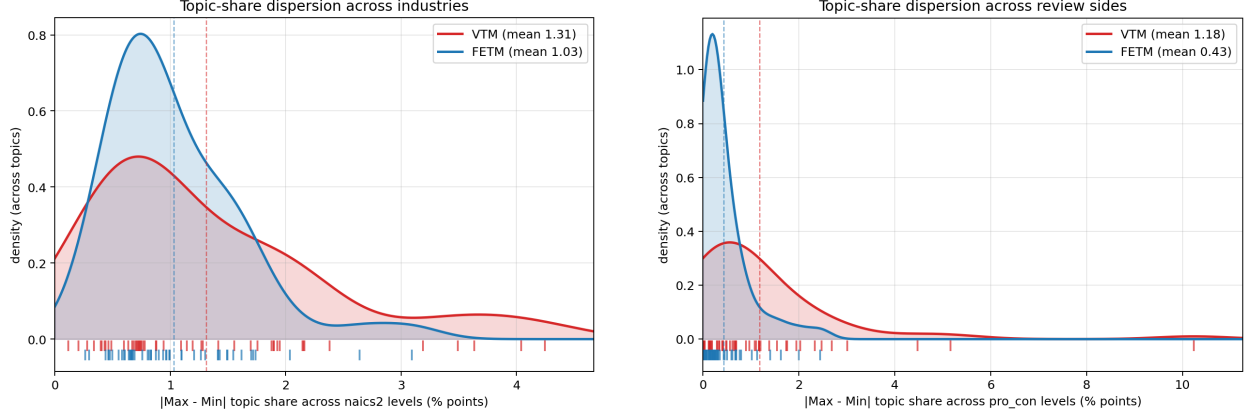


Figure 4: Kernel density estimates of the per-topic spread Δ_k at $K = 50$. Left: industries, where the reduction under mFETM is not statistically distinguishable at the 5% level. Right: pro/con, where the reduction is larger and statistically significant.

Table 5: Topic-share dispersion across environments (mFETM vs. mVTM, $K = 50$)

	Industries	Pro / Con
mVTM mean (% pts)	1.31	1.18
mFETM mean (% pts)	1.03	0.43
Difference (mFETM – mVTM)	-0.28*	-0.75***
	(0.16)	(0.25)
Welch t statistic	-1.72	-2.98
p -value (two-sided)	0.090	0.004
N (topics)	50	50

Notes. Each observation is a topic. “Max–Min” is the difference between the largest and smallest mean topic share across the levels of the column variable (NAICS-2 industries or Pro / Con). Topics are not aligned across models, so the test treats the two sets of topic-level values as independent samples (Welch two-sample t). Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. Held-out test split.

One topic share is omitted because the K shares sum to one. Results are in Table 6. mVTM topics explain more than half of review polarity ($R^2 = 0.554$): knowing its topic mixture goes a long way toward revealing whether the text came from the positive or negative side. Under mFETM, that figure falls to 0.197. The estimated topic measures themselves contain far less information about review tone.

We then ask whether the topics predict industry, one binary contrast per sector pair across eight selected NAICS-2 sectors (Table 7). mVTM has the higher R^2 in all 28 contrasts. Regardless of which two sectors are compared, the conventional topic vector reveals more about industry. The mVTM-to-mFETM R^2 ratio reaches $2.4\times$ for Manufacturing versus Finance and remains $1.09\times$ even for the closest comparison, Information versus Professional Services.

The two exercises agree. mFETM topics move less across environments, and the full topic vector makes those environments harder to predict, while the topics keep their substantive workplace content. The next section uses the topics to characterize top employers.

Table 6: Predicting Pro / Con Review Side from Topic Shares ($K = 50$)

Contrast	N	mFETM R^2	mVTM R^2	mV – mF	mV / mF
Pro vs Con	906,938	0.197	0.554	0.356	$2.8\times$

Notes. OLS of the pro/con indicator on an intercept and $K - 1$ topic shares; the final share is omitted because the shares sum to one. The last two columns report mVTM minus mFETM and their ratio. Full corpus.

Table 7: Predicting Industry from Topic Shares ($K = 50$)

Industry contrast	N	mFETM R^2	mVTM R^2	mV – mF	mV / mF
Finance vs Accom & Food	178,456	0.210	0.283	0.073	$1.3\times$
Accom & Food vs Health Care	86,546	0.094	0.159	0.066	$1.7\times$
Retail vs Finance	245,677	0.125	0.185	0.060	$1.5\times$
Information vs Accom & Food	221,425	0.227	0.286	0.058	$1.3\times$
Accom & Food vs Prof Services	98,954	0.238	0.294	0.057	$1.2\times$
Accom & Food vs Real Estate	90,384	0.135	0.191	0.056	$1.4\times$
Information vs Retail	288,646	0.154	0.206	0.052	$1.3\times$
Manufacturing vs Accom & Food	236,446	0.242	0.287	0.044	$1.2\times$
Retail vs Health Care	153,767	0.037	0.078	0.041	$2.1\times$
Manufacturing vs Finance	274,808	0.026	0.061	0.035	$2.4\times$
Real Estate vs Health Care	36,836	0.040	0.074	0.035	$1.9\times$
Manufacturing vs Retail	303,667	0.179	0.211	0.032	$1.2\times$
Prof Services vs Health Care	45,406	0.096	0.126	0.030	$1.3\times$
Manufacturing vs Health Care	182,898	0.041	0.069	0.029	$1.7\times$
Retail vs Prof Services	166,175	0.099	0.125	0.026	$1.3\times$
Information vs Health Care	167,877	0.039	0.060	0.021	$1.5\times$
Finance vs Health Care	124,908	0.024	0.044	0.020	$1.8\times$
Retail vs Real Estate	157,605	0.051	0.070	0.020	$1.4\times$
Information vs Finance	259,787	0.020	0.037	0.017	$1.9\times$
Retail vs Accom & Food	207,315	0.057	0.073	0.016	$1.3\times$
Finance vs Real Estate	128,746	0.013	0.027	0.014	$2.1\times$
Manufacturing vs Real Estate	186,736	0.021	0.033	0.011	$1.5\times$
Manufacturing vs Information	317,777	0.017	0.029	0.011	$1.6\times$
Information vs Real Estate	171,715	0.019	0.030	0.010	$1.5\times$
Finance vs Prof Services	137,316	0.026	0.035	0.009	$1.4\times$
Prof Services vs Real Estate	49,244	0.053	0.062	0.009	$1.2\times$
Manufacturing vs Prof Services	195,306	0.012	0.021	0.009	$1.7\times$
Information vs Prof Services	180,285	0.015	0.016	0.001	$1.1\times$

Notes. Each row is a separate OLS of an industry indicator on an intercept and $K - 1$ topic shares. The final share is omitted because the shares sum to one. The last two columns report mVTM minus mFETM and their ratio. Full corpus.

6 Topics and Top Employers

We next ask whether the less environment-associated topics tell us something useful about corporate culture. Fortune-list inclusion provides an external benchmark of employer quality. Beyond fit, the exercise shows what each model treats as a meaningful difference between firms. A topic model dominated by review polarity may predict the list simply by detecting complaints. A model that separates polarity and industry language from shared content can instead reveal the underlying workplace dimensions that distinguish top employers.

The outcome is $y_{f,t} = 1$ if firm f appears on Fortune’s 100 Best Companies to Work For list in year t . The lists for 2009–2022 contain 1,400 firm-year slots covering 328 published company names. Fuzzy matching is used only to generate candidates. Every candidate is then manually accepted or rejected; the reviewed crosswalk accepts 95 Fortune names, covering 76 Glassdoor firm IDs and 425 list-year slots. The remainder are largely private companies outside the S&P 1500 panel or names without a defensible link.

Keeping firm-year cells with at least five documents yields a panel of 7,992 cells across 911 firms and fourteen years, constructed from 863,501 pro/con documents. The median cell contains 28 documents, and 323 cells (4.0%) are on the list.

Topic features are the within-cell means $\bar{\pi}_{f,t,k}^m$, computed separately for mFETM and mVTM at $K = 50$. For each model, we fit

$$\Pr(y_{f,t} = 1) = \Lambda\left(\alpha_t^m + \sum_k \delta_k^m \bar{\pi}_{f,t,k}^m\right),$$

with year fixed effects α_t^m and standard errors clustered at the firm.⁴

To give every topic the same reference, we report

$$\Delta_k^m = \delta_k^m - \frac{1}{K-1} \sum_{j \neq k} \delta_j^m.$$

This is the log-odds effect of shifting topic-share mass toward topic k while taking equal amounts from every other topic. Its contrast weights sum to zero, so the estimate and its firm-clustered standard error are invariant to the topic omitted during estimation. Positive values indicate association with Fortune-list inclusion relative to the average of the other topics; negative values indicate the reverse.

6.1 Which Topics Predict Fortune-List Inclusion?

This exercise gives the environment-agnostic property a substantive payoff. Because mFETM topics are less tied to industry and review side, their firm-level associations can be read as dimensions of

⁴The K topic shares sum to one, so one topic is omitted internally to obtain a full-rank design. This choice does not affect fitted values or the normalization-invariant contrasts reported below.

corporate culture rather than primarily as markers of sectoral vocabulary or positive and negative tone.

Each model yields 50 normalization-invariant topic contrasts. We select main-table rows using the within-model Benjamini–Hochberg (BH) procedure at a 5% false discovery rate. Stars report unadjusted two-sided p -values; they do not select rows. The complete estimates, including raw p -values and BH-adjusted q -values, are reported in the replication materials.

Table 8 reports the nine mFETM topics that survive this adjustment. Positive contrasts include compensation and benefits, work-life balance, market positioning, general workplace experience, and amenities. Negative contrasts include restructuring and layoffs, abusive management, benefit packages, and coworker friendliness. A negative contrast does not mean that friendly coworkers or benefits are intrinsically undesirable: it means that shifting share toward that topic, rather than equally toward the other topics, is associated with lower Fortune-list inclusion. Taken together, the estimates map corporate culture along compensation, flexibility, collegiality, management quality, job security, and the workplace environment, rather than along positive or negative language.

At the same threshold, mVTM identifies seven topics (Table 9). These include flexible work, food and beverage benefits, and technology strategy, alongside workload pressure, poor management, retaliatory management, and workforce reductions. Technology strategy is also among mVTM’s more industry-associated topics. Its presence illustrates the remaining confounding: mVTM mixes general workplace attributes with signals about the environments in which firms operate, so its map of what makes a firm a good employer is less complete and less portable. A more discriminating comparison asks whether those associations survive the removal of topics tied most strongly to industry and review polarity. We turn to it next.

Table 8: Fortune top-100 topic predictors, mFETM ($K = 50$). Each coefficient contrasts one topic with the equally weighted average of all other topics and is therefore invariant to the fitting baseline. Rows pass within-model BH $q \leq 0.05$ across all 50 contrasts. The firm-year logit includes year FE and firm-clustered SEs. Stars use raw two-sided p -values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Topic (mFETM)	$\hat{\Delta}_k$ (SE)	BH q
26. Employee compensation and benefits	+39.89 (8.05)***	0.000
27. Company market positioning and competitiveness	+31.31 (7.22)***	0.000
10. Work-life balance and flexibility	+27.78 (6.53)***	0.000
8. Workplace Job Experience Evaluation	+14.06 (4.68)***	0.019
15. Employee perks and workplace amenities	+11.57 (4.26)***	0.037
24. Employee benefits packages	-14.40 (5.16)***	0.033
2. Abusive management and poor working conditions	-33.75 (10.29)***	0.009
37. Shareholder-driven restructuring and layoffs	-46.58 (8.41)***	0.000
35. Coworker friendliness and collegiality	-77.38 (14.87)***	0.000
N = 7,992; year FE; firm-clustered SEs; pseudo- $R^2 = 0.1584$; $ z \geq 1.96$ and BH $q \leq 0.05$ topics only		

Table 9: Fortune top-100 topic predictors, mVTM ($K = 50$). The normalization-invariant contrast, selection rule, specification, and stars follow Table 8.

Topic (mVTM)	$\hat{\Delta}_k$ (SE)	BH q
5. Employee food and beverage benefits	+20.15 (5.35)***	0.003
6. Enterprise technology strategy and innovation	+16.73 (5.94)***	0.035
10. Flexible work arrangements and remote work	+15.25 (5.17)***	0.026
41. Workplace stress and workload pressure	+11.06 (3.75)***	0.026
28. Abusive and retaliatory management culture	-17.61 (5.93)***	0.026
15. Market downturns and workforce reductions	-26.54 (5.41)***	0.000
2. Poor management practices	-46.69 (10.12)***	0.000

N = 7,992; year FE; firm-clustered SEs; pseudo- $R^2 = 0.1363$; $|z| \geq 1.96$ and BH $q \leq 0.05$ topics only

6.2 Comparing mFETM and mVTM

With all topics included, mFETM fits the Fortune data slightly better: its pseudo- R^2 is 0.158, compared with 0.136 for mVTM. We then ask where that fit comes from. To do so, we remove each model’s topics that are most associated with industry or review polarity and re-estimate the Fortune model. If mVTM relies more heavily on environment-specific topics, its fit should deteriorate faster.

Figure 5 reports the full removal path. After removing 10% of topics, mFETM retains 96% of its original fit, compared with 84% for mVTM. After removing 30%, the corresponding figures are 89% and 54%. mFETM’s ability to distinguish top employers is therefore distributed across general workplace themes, whereas much of mVTM’s fit is concentrated in topics that also encode industry or review tone.

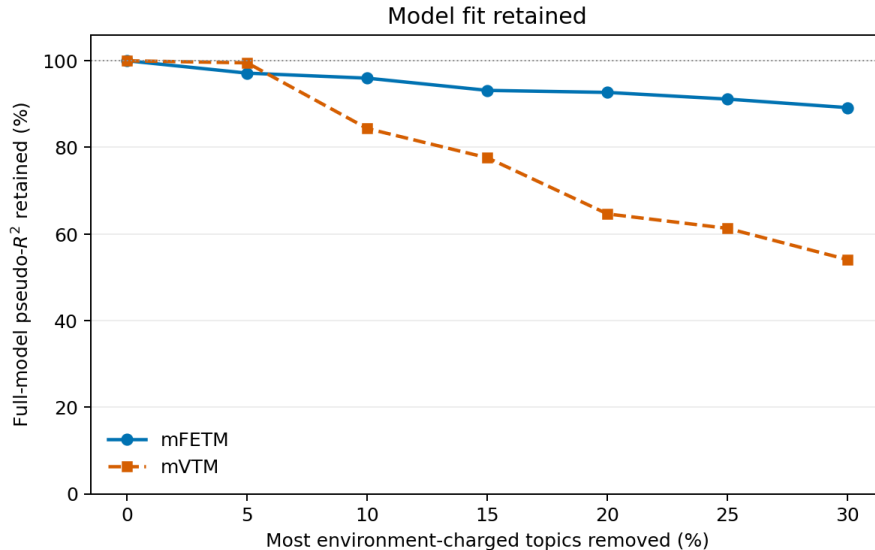


Figure 5: Model fit retained after removing the topics most associated with industry or review polarity, $K = 50$. Each line reports McFadden pseudo- R^2 as a percentage of that model’s full fit. Appendix H gives the ranking and cutoff construction.

Appendix H reports the corresponding Vuong comparison, random-removal benchmark, and polarity-only topic-by-topic path.

7 Steering Along Estimated Fixed-Effect Directions

So far, we have used mFETM to recover topics that are less tied to industry and review polarity while retaining predictive and substantive information. We now use the model’s separately estimated, topic-specific environment directions. These directions let us hold a topic fixed and change how it is expressed across review sides or industries. Reading the generated text shows what each fixed effect encodes, and the same construct can be translated across contexts or used to generate controlled examples.

Each retained SAE concept i has a decoder direction $\mathbf{W}_{\text{dec}}[i] \in \mathbb{R}^{3584}$. For topic k and environment level e , mFETM implies the topic-conditional concept distribution

$$p_{e,k} = \text{softmax}(\beta_k^* + \gamma_{e,k}).$$

Its decoder-weighted average defines a composite direction,

$$\mathbf{s}_{e,k} = \frac{\sum_i p_{e,k,i} \mathbf{W}_{\text{dec}}[i]}{\|\sum_i p_{e,k,i} \mathbf{W}_{\text{dec}}[i]\|_2}.$$

To compare two levels a and b of the same fixed-effect block, we difference their fitted concept distributions before mapping them through the decoder:

$$\mathbf{s}_{a-b,k} = \frac{\sum_i (p_{a,k,i} - p_{b,k,i}) \mathbf{W}_{\text{dec}}[i]}{\|\sum_i (p_{a,k,i} - p_{b,k,i}) \mathbf{W}_{\text{dec}}[i]\|_2}.$$

The subtraction removes the topic’s common component and retains the fitted shift between the two environments. Positive steering along $\mathbf{s}_{a-b,k}$ moves toward level a ; negative steering moves toward level b . The other fixed-effect block remains at its centered mean, so the comparison varies one fitted refinement at a time.

We implement the intervention following Zheng et al. (2025). Let $\mathbf{a}$ denote Gemma’s incoming residual-stream activation, $\mathbf{b}$ the SAE decoder bias, and $\mathbf{s}$ the selected direction. At decoder blocks 40 and 41 we remove the activation’s existing projection onto $\mathbf{s}$, replace it with strength λ , and restore the bias:

$$\bar{\mathbf{a}} = \mathbf{a} - \mathbf{b}, \quad \bar{\mathbf{a}}_{\perp} = \bar{\mathbf{a}} - (\bar{\mathbf{a}} \cdot \mathbf{s})\mathbf{s}, \quad \mathbf{a}_{\text{steered}} = \bar{\mathbf{a}}_{\perp} + \lambda \mathbf{s} + \mathbf{b}.$$

Gemma’s weights never change. Within each comparison, the topic, review prompt, and generation procedure are held fixed; only the estimated environment direction and its sign vary.

The exercise holds one topic fixed throughout. We use mFETM Topic 8, “Workplace Job Experience Evaluation,” because it is one of the model’s broader workplace topics and can naturally be

expressed across both review sides and industries. The same construction applies to any estimated topic. Holding the topic fixed ensures that differences across continuations come from the estimated environment refinement rather than from changing what the review is about.

We run two within-topic comparisons. The first isolates review polarity. Starting from the same review, we steer toward the Pro and Con poles of the fitted Pro-minus-Con direction. This asks whether mFETM can change how a common workplace experience is evaluated while preserving its subject. The second isolates industry expression. We steer the same topic and review toward Accommodation and Food Services (NAICS 72) and Information (NAICS 51). This asks whether the model can translate a common workplace theme into the vocabulary and setting associated with two different sectors. Every continuation begins from the same prompt (“Fun atmosphere. Worked very independently. A lot of really great people at the company. The industry has its pros and cons.”) and uses $|\lambda| = 400$ so that the fitted contrasts are clearly visible in generated language.

Table 10: Generated continuations along estimated fixed-effect directions.

Common prompt. “Fun atmosphere. Worked very independently. A lot of really great people at the company. The industry has its pros and cons.”

Panel A. Review polarity

Pro

$\lambda = +400$

The job was rewarding because I got to work with a diverse group of employees from all over the world who were passionate about their jobs; however it also had some challenges such as long hours which made me feel like my life experience would be enriched by working there for an extended period time (which is why this position ended up being one that lasted 10 years).

Con

$\lambda = -400$

The work was difficult but I learned a ton about how to get things done quickly with little resources (time/money). It’s been hard for me because it requires more creativity than what my current job does which isn’t as much creative thinking involved either way so that might be why this seems harder sometimes too?

Panel B. Industry

**Accommodation and
Food Services**

NAICS 72, $\lambda = +400$

The customer service was good but not as fun to work with because they were a bit more stressful than some other customers I had worked with before during my time there (I am sure this is due mostly on how busy it gets). Overall though overall experience would be 5/10 if you’re looking for something relaxing after school hours then definitely check out their website!

Information

NAICS 51, $\lambda = -400$

The corporate office was a big plus for me as I am an out going person who likes to be around other companies/people all day long instead of sitting alone behind my desk every single minute!

Notes. Topic 8 and the review prompt are fixed across all rows. Within each panel, the two rows use opposite signs of the same estimated contrast direction at $|\lambda| = 400$. The intervention is applied at decoder blocks 40–41.

Table 10 makes the two refinements readable. In Panel A, the Pro pole expresses the fixed workplace-experience topic through reward, diversity, and employees’ commitment; the Con pole expresses it through difficulty, limited resources, and communication problems. The review remains about workplace experience, but its evaluative framing changes. In Panel B, the NAICS 72 pole

moves the same topic toward customer service and a busy service setting, whereas the NAICS 51 pole moves it toward a corporate-office setting. Here the contextual vocabulary changes while the underlying theme does not. Because both the topic and starting review are fixed, these comparisons isolate how mFETM represents polarity and industry within a common semantic object. A conventional topic model cannot execute this comparison because it has no separately estimated environment direction: contextual language remains bundled into the topic itself.

We next test whether the Pro–Con direction moves Gemma in the intended direction beyond this single example. We fix Topics 1–10 before inspecting the results and use the same balanced set of ten Pro and ten Con documents from the held-out test split for every topic and every value of $\lambda \in \{-400, -300, \dots, 400\}$. For each topic and strength, we compute the difference between the mean per-token log likelihood of the Pro and Con documents and express it as a lift from $\lambda = 0$. Thus, a rising curve means that positive steering makes Pro reviews relatively more likely and negative steering makes Con reviews relatively more likely. The mean lift rises almost linearly from -0.544 at $\lambda = -400$ to 0.514 at $\lambda = 400$; nine of ten topic directions move correctly at both endpoints.

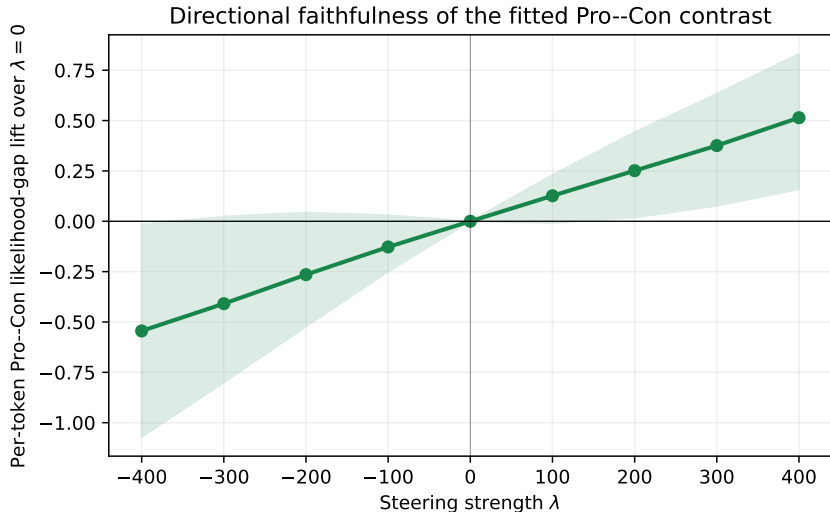


Figure 6: Directional faithfulness of the fitted Pro-minus-Con contrast. The line is the mean likelihood-gap lift across ten topics fixed in advance; the band spans the 10th to 90th percentiles across topics. Every topic-specific curve is centered at $\lambda = 0$, which is the direction-specific ablation.

The generated examples and likelihood path serve complementary purposes. The examples show what the estimated fixed effects mean in language; the likelihood path establishes that the Pro–Con direction operates systematically rather than only for the displayed prompt. Appendix G reports the accompanying fluency check. The fixed effects thus reveal how a common workplace theme is translated across review sides and sectors, and they provide executable, topic-specific directions for producing or recognizing those contextual expressions.

8 Conclusion

Topic models are useful for comparative research only when the same topic has the same meaning across documents. That requirement is difficult to satisfy in multi-source corpora: a common theme may be expressed differently across environments, while word counts can miss concepts carried by phrases, context, and tone. The mechanistic Fixed-Effects Topic Model addresses both problems. It represents documents through interpretable concepts extracted from a language model and separates each topic’s global content from the environment-specific language used to express it.

The empirical results show that this separation works. mFETM improves held-out document-completion perplexity while producing substantially more portable topic measures. mVTM’s topic shares explain 55.4% of review polarity, compared with 19.7% for mFETM, and predict industry better in all 28 reported pairwise sector comparisons. The fixed effects therefore remove a large amount of contextual variation from the measures researchers would use to compare firms.

That portability has a substantive payoff. Normalization-invariant Fortune coefficients give mFETM a broader, more interpretable view of employer quality, spanning compensation, work-life balance, coworker relations, management, job security, and workplace amenities. mVTM also contains significant topic associations, so their number alone is not the main comparison. Its inclusion of industry-linked technology strategy illustrates why its substantive picture is narrower. The sharper result concerns where the models obtain their fit. After removing the topics most associated with industry and review polarity, mFETM retains substantially more of its original fit throughout the 10%–30% removal range. Its firm-level information is spread across general workplace themes rather than concentrated in environment-specific topics.

Finally, the mechanistic representation turns the estimated objects into directions inside the language model. Holding a topic fixed, we can steer along its estimated industry or review-side refinement and generate how that same theme is expressed in a different environment. The fixed effects are therefore interpretable and manipulable objects as well as statistical controls. This combination of comparability, conceptual interpretation, and controlled generation extends beyond employee reviews to any corpus in which shared constructs are expressed differently across sources, including political text across outlets and earnings calls across sectors.

References

- Bandiera, Oriana, Andrea Prat, Stephen Hansen, and Raffaella Sadun (2020). “CEO behavior and firm performance”. In: *Journal of Political Economy* 128.4, pp. 1325–1369.
- Bills, Steven, Nick Cammarata, Dan Mossing, Henk Tillman, Leo Gao, Gabriel Goh, Ilya Sutskever, Jan Leike, Jeff Wu, and William Saunders (2023). “Language Models Can Explain Neurons in Language Models”. In: *OpenAI Blog*.
- Blei, David M, Alp Kucukelbir, and Jon D McAuliffe (2017). “Variational inference: A review for statisticians”. In: *Journal of the American statistical Association* 112.518, pp. 859–877.
- Blei, David M, Andrew Y Ng, and Michael I Jordan (2003). “Latent dirichlet allocation”. In: *Journal of machine Learning research* 3.Jan, pp. 993–1022.
- Bricken, Trenton, Adly Templeton, Joshua Batson, et al. (2023). “Towards Monosemanticity: Decomposing Language Models with Dictionary Learning”. In: *Transformer Circuits Thread*.
- Cai, Wei, Andrea Prat, and Jiehang Yu (2024). “Measuring Organizational Capital”. In: *Available at SSRN 4870532*.
- Cameron, Kim S. and Robert E. Quinn (2011). *Diagnosing and Changing Organizational Culture: Based on the Competing Values Framework*. 3rd ed. San Francisco: Jossey-Bass.
- Card, Dallas, Chenhao Tan, and Noah A. Smith (2018). “Neural Models for Documents with Metadata”. In: *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 2031–2040.
- Cunningham, Hoagy, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey (2023). “Sparse Autoencoders Find Highly Interpretable Features in Language Models”. In: *arXiv preprint arXiv:2309.08600*.
- Eisenstein, Jacob, Amr Ahmed, and Eric P. Xing (2011). “Sparse Additive Generative Models of Text”. In: *Proceedings of the 28th International Conference on Machine Learning (ICML)*, pp. 1041–1048.
- Elhage, Nelson, Tristan Hume, Catherine Olsson, et al. (2022). “Toy Models of Superposition”. In: *Transformer Circuits Thread*.
- Gemma Team (2024). “Gemma 2: Improving Open Language Models at a Practical Size”. In: *arXiv preprint arXiv:2408.00118*.
- Gentzkow, Matthew, Bryan Kelly, and Matt Taddy (2019). “Text as data”. In: *Journal of Economic Literature* 57.3, pp. 535–574.
- Gibbons, Robert and Rebecca Henderson (2013). “What do managers do? Exploring persistent performance differences among seemingly similar enterprises”. In: *The Handbook of Organizational Economics*. Ed. by Robert Gibbons and John Roberts. Princeton University Press, pp. 680–731.
- Green, Thomas C., Russell Jame, and Brandon Lock (2019). “Crowdsourced analyses of corporate culture and firm performance”. In: *Journal of Financial Economics* 133.3, pp. 676–697.
- Grimmer, Justin and Brandon M Stewart (2013). “Text as data: The promise and pitfalls of automatic content analysis methods for political texts”. In: *Political analysis* 21.3, pp. 267–297.

- Guiso, Luigi, Paola Sapienza, and Luigi Zingales (2015). “The Value of Corporate Culture”. In: *Journal of Financial Economics* 117.1, pp. 60–76.
- Handy, Charles (1978). *Gods of Management: The Changing Work of Organizations*. Oxford: Oxford University Press.
- Hansen, Stephen, Michael McMahon, and Andrea Prat (2018). “Transparency and deliberation within the FOMC: A computational linguistics approach”. In: *The Quarterly Journal of Economics* 133.2, pp. 801–870.
- Karabarbounis, Loukas and Gabriel Pinto (2018). *The Macroeconomic Effects of Firm-Level Uncertainty: A Case Study of an Online Review System*. Tech. rep. w25090. National Bureau of Economic Research.
- Kingma, Diederik P and Jimmy Ba (2014). “Adam: A method for stochastic optimization”. In: *arXiv preprint arXiv:1412.6980*.
- Kingma, Diederik P and Max Welling (2013). “Auto-encoding variational Bayes”. In: *arXiv preprint arXiv:1312.6114*.
- Kotter, John P. and James L. Heskett (2011). *Corporate Culture and Performance*. New York: Free Press.
- Lee, Jihun, Scott D. Dyreng, Jeffrey L. Hoopes, and Jaron H. Wilde (2021). “The effects of corporate tax avoidance news on employee perceptions of their employers”. In: *Journal of Business Ethics* 171.4, pp. 749–771.
- Li, Kai, Feng Mai, Rui Shen, and Xinyan Yan (2021). “Measuring Corporate Culture Using Machine Learning”. In: *The Review of Financial Studies* 34.7, pp. 3265–3315.
- Lieberum, Tom, Senthooan Rajamanoharan, Arthur Conmy, et al. (2024). “Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on Gemma 2”. In: *arXiv preprint arXiv:2408.05147*.
- Marinescu, Ioana, Sanjid Islam, and Ellis J. G. Scharfenaker (2021). “The incentives to review: Evidence from Glassdoor.com”. In: *Labour Economics* 73, p. 102076.
- Quinn, Kevin M, Burt L Monroe, Michael Colaresi, Michael H Crespin, and Dragomir R Radev (2010). “How to analyze political attention with minimal assumptions and costs”. In: *American Journal of Political Science* 54.1, pp. 209–228.
- Rajamanoharan, Senthooan, Tom Lieberum, Nicolas Sonnerat, et al. (2024). “Jumping Ahead: Improving Reconstruction Fidelity with JumpReLU Sparse Autoencoders”. In: *arXiv preprint arXiv:2407.14435*.
- Ranganath, Rajesh, Sean Gerrish, and David Blei (2014). “Black box variational inference”. In: *Artificial intelligence and statistics*. PMLR, pp. 814–822.
- Roberts, Margaret E, Brandon M Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G Rand (2014). “Structural topic models for open-ended survey responses”. In: *American journal of political science* 58.4, pp. 1064–1082.
- Schein, Edgar H. (1985). *Organizational Culture and Leadership*. San Francisco: Jossey-Bass.

- Sim, Yanchuan, Bryan Routledge, and Noah Smith (2015). “The utility of text: the case of Amicus briefs and the Supreme Court”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 29. 1.
- Syverson, Chad (2004). “Market structure and productivity: A concrete example”. In: *Journal of Political Economy* 112.6, pp. 1181–1222.
- Vuong, Quang H. (1989). “Likelihood Ratio Tests for Model Selection and Non-Nested Hypotheses”. In: *Econometrica* 57.2, pp. 307–333.
- Zheng, Carolina, Nicolas Beltran-Velez, Sweta Karlekar, Claudia Shi, Achille Nazaret, Asif Mallik, Amir Feder, and David M. Blei (2025). “Model directions, not words: Mechanistic topic models using sparse autoencoders”. In: *arXiv preprint arXiv:2507.23220*.

A Additional Introduction Tables

This appendix documents the three steps behind the introductory Amazon–Aflac example. Table A1 reports the reviews’ leading topics under the historical bag-of-words models; Table A2 contrasts that representation with the current bag of concepts; and Table A3 displays the concept side of a naming procedure that also uses representative reviews. The historical estimates motivate the use of fixed effects but are not directly comparable to the current $K = 50$ models.

Table A1: Leading Topic Shares under Historical Bag-of-Words Models

Review	Model	Top Words	Share	
Amazon	VTM	managers, industry, products, communication, organization, awesome, highly, cost, provide	0.24	
		told, generous, firm, executives, timings, left, directors, saying, defined	0.15	
		fun, days, companies, flexibility, associates, plenty, clients, daily, treat	0.08	
		lack, home, cons, family, bonus, issues, processes, cool, quickly	0.08	
	FETM	overall, looking, free, left, chance, workplace, reason, lunch, food	0.38	
		focus, major, follow, groups, encouraged, teams, feeling, executives, credit	0.12	
		based, department, job, given, performance, managers, supervisors, truly, help	0.12	
		jobs, customer, person, issues, problems, treat, hiring, possible, drive	0.08	
	Aflac	VTM	training, high, changes, performance, stressful, constant, turnover, layoffs, lunch	0.16
			poor, decent, free, workers, insurance, shift, skills, world, helpful	0.12
told, generous, firm, executives, timings, left, directors, saying, defined			0.08	
bad, focus, pressure, especially, discounts, ibm, services, phone, pays			0.07	
FETM		overall, looking, free, left, chance, workplace, reason, lunch, food	0.29	
		jobs, customer, person, issues, problems, treat, hiring, possible, drive	0.14	
		opportunities, advancement, career, growth, development, professional, willing, technology, systems	0.10	
		learning, diversity, stock, takes, constantly, options, promote, tools, point	0.07	

Notes. Historical $K = 30$ bag-of-words illustration archived in the earlier FETM draft. VTM denotes the historical vanilla topic model and FETM its fixed-effects extension; neither is the current mechanistic mVTM or mFETM. These estimates are retained to document the motivating example and are not directly comparable to the current $K = 50$ results.

Table A2: From Words to Concepts in the Introductory Example

Bag of words: top words	Bag of concepts: current topic
Amazon	
<i>overall, looking, free, left, chance, workplace, reason, lunch</i>	Warehouse Operations and Equipment Handling: technical terms related to online support services; phrases and terms related to automated processes and claims management; terms related to home appliance service and repair
<i>focus, major, follow, groups, encouraged, teams, feeling, executives</i>	Attendance and time-off policies: terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); phrases indicating tasks and responsibilities; mentions of job performance and employment situations
<i>based, department, job, given, performance, managers, supervisors, truly</i>	Wages and Working Hours: terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); mentions of job performance and employment situations; quantitative data and time metrics related to events or activities
Aflac	
<i>overall, looking, free, left, chance, workplace, reason, lunch</i>	Attendance and time-off policies: terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); phrases indicating tasks and responsibilities; mentions of job performance and employment situations
<i>jobs, customer, person, issues, problems, treat, hiring, possible</i>	Warehouse Operations and Equipment Handling: technical terms related to online support services; phrases and terms related to automated processes and claims management; terms related to home appliance service and repair
<i>opportunities, advancement, career, growth, development, professional, willing, technology</i>	Variable work schedules and hours: mentions of scheduling appointments and time management; terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); mentions of job performance and employment situations

Notes. For each review, the rows are its three leading topics, paired by rank for exposition rather than aligned as latent objects. Left: top words from the historical bag-of-words FETM in Table A1. Right: the current mFETM topic name followed by its three highest-ranked retained concept descriptions. The current mFETM estimates give the two reviews different leading topics.

Table A3: From Concepts to Names: the Interpretation Step

Top concepts	Topic name
technical terms related to online support services; phrases and terms related to automated processes and claims management; terms related to home appliance service and repair; references to order-picking machinery and their operational details	Warehouse Operations and Equipment Handling
terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); phrases indicating tasks and responsibilities; mentions of job performance and employment situations; references to past events or experiences	Attendance and time-off policies
terms related to employee rights and leaves of absence under the Family Medical Leave Act (FMLA); mentions of job performance and employment situations; quantitative data and time metrics related to events or activities; various forms of currency and mentions of monetary values	Wages and Working Hours

Notes. Three topics from the example. Left: the highest-weighted concepts under $\text{softmax}(\beta_k^*)$, given by the features’ automatic descriptions. Right: the generated name, which also uses representative high-share reviews as described in Appendix D. Names label the output; estimation never uses them.

B Estimated Topic Names

Table A4 lists the language-model-generated names of all $K = 50$ topics. Candidate concepts are drawn from $\text{softmax}(\beta_k^*)$ for mFETM and $\text{softmax}(\beta_k)$ for mVTM; both models then use the same refinement and review-grounding procedure described in Appendix D. Topic numbers are model-specific, so rows are not aligned across columns.

Table A4: Estimated topic names at $K = 50$.

mFETM		mVTM	
1	Employee compensation and career development	Employee benefits and career growth opportunities	
2	Abusive management and poor working conditions	Poor management practices	
3	Enterprise project management and integration	Supportive management and coworkers	
4	Attendance and time-off policies	Employee autonomy and growth opportunities	
5	Management care for employees	Employee food and beverage benefits	
6	Leadership and organizational culture	Enterprise technology strategy and innovation	
7	Entry-level hiring and career opportunities	Positive workplace experience	
8	Workplace Job Experience Evaluation	Overwork and workplace stress	
9	Supportive Coworker Relationships	Workplace management and support	
10	Work-life balance and flexibility	Flexible work arrangements and remote work	
11	Workplace management fairness and transparency	Employee compensation and career advancement	
12	Employee discounts and sales incentives	Manual material handling strain	
13	Organizational change adaptability	Workplace communication and cooperation	
14	Warehouse Operations and Equipment Handling	Call center workload and performance monitoring	
15	Employee perks and workplace amenities	Market downturns and workforce reductions	
16	Competitive pay and steady work hours	Positive fast-paced work environment	
17	Variable work schedules and hours	Employee salary competitiveness	
18	Positive workplace culture and environment	Workplace culture and employee experience	
19	Wages and Working Hours	Employee well-being and safety culture	
20	Workplace environment and culture	Company reputation and industry leadership	
21	Retail staffing and management conditions	Workload and management practices	
22	Organizational values alignment and integrity	Organizational culture and leadership development	
23	Unethical management and unfair advancement	Politicized, exclusionary workplace culture	
24	Employee benefits packages	Organizational culture and values	
25	Workplace workload and work-life balance	Forced-ranking performance evaluations	
26	Employee compensation and benefits	Employee compensation and workload conditions	
27	Company market positioning and competitiveness	Entry-level hiring and advancement criteria	
28	Coworker quality and collaboration	Abusive and retaliatory management culture	
29	Organizational culture and management practices	Organizational bureaucracy and decision-making delays	
30	Telecommunications company offerings and employee benefits	Retail staffing and management problems	
31	Work Effort and Advancement	Workload and work-life balance	
32	Employee pay and salary levels	Employee benefits package	
33	Career advancement opportunities	Career advancement opportunities	
34	Career advancement opportunities	Job security and advancement pressure	
35	Coworker friendliness and collegiality	High-pressure sales quotas and product pushing	
36	Positive employee experience and workplace relationships	Variable work schedules and limited flexibility	
37	Shareholder-driven restructuring and layoffs	Workplace fairness and advancement opportunities	
38	Workplace location and physical facilities	Absence of reported workplace drawbacks	
39	Workplace management quality and conditions	Positive coworker relationships and caliber	
40	Workplace management and organizational culture	Low pay and compensation competitiveness	
41	Employee compensation and benefits	Workplace stress and workload pressure	
42	Positive workplace sentiment	Workplace location and commuting conditions	
43	Workplace location and relocation tradeoffs	Workplace compensation and scheduling conditions	
44	Industry downturn and market volatility	Employee training and skill development	
45	Performance evaluation and career advancement	Management quality and leadership effectiveness	
46	Fast-paced workplace culture	Abundance of workplace opportunities	
47	Poor management and unfair treatment	Positive workplace culture	
48	Poor management and workplace support	Pay, hours, and scheduling conditions	
49	No reported workplace issues	Workplace facilities and amenities	
50	Employee training and professional development	Compensation and senior management priorities	

C Bag-of-Concepts Construction Details

Each document is truncated at 256 tokens and passed through the base Gemma-2-9B model. We read the residual stream at layer 39 (3,584 dimensions) and encode each token’s activation with the width-16,384 JumpReLU autoencoder (Rajamanoharan et al., 2024) from the Gemma Scope suite (Lieberum et al., 2024). The SAE writes each activation as a sparse, nonnegative vector of feature firings, and we keep the nonzero (document, token, feature, magnitude) entries.

Three pruning rules follow:

1. before applying the strength threshold q_v , drop features with any positive activation in fewer than 0.1% of documents, which are too rare to learn from, or more than 90%, which are too common to discriminate;
2. drop features active on more than 1% of tokens in the SAE’s training data, excluding ubiquitous low-level features that would otherwise create “syntax topics”;
3. show each remaining feature’s automatic label to GPT-4o-mini and drop descriptions of programming, mathematics, grammar, text formatting, or stop words (prompt in Appendix D).

The three rules reduce the dictionary from 16,384 candidate features to 6,889. For each remaining feature, the firing threshold q_v is the 80th percentile of its nonzero activation distribution in the SAE’s training data, computed from Neuronpedia’s per-feature histogram.⁵ The implementation uses the corresponding corpus nonzero-activation percentile when a histogram is unavailable, although every feature in the final run had a histogram. After thresholding, we again require positive counts in at least 0.1% of documents, leaving 4,735 features.

The supplied Neuronpedia feature labels describe activations in general web text, which may not match their meaning in workplace reviews; we note the divergences where they matter. The resulting count matrix is $906,938 \times 4,735$ and about 2.4% dense. Both models use this same matrix, fit/validation/test partition, topic count, and encoder architecture; their separately selected optimization schedules are reported in the main text.

D LLM Prompts and Details

Three steps use a language model: filtering spurious SAE features (Section 2.3), removing out-of-place features from a topic’s representation, and naming topics (Section 4.1). We use fixed prompts and model versions and hold the resulting outputs fixed across analyses. This appendix reports the models and substantive prompt instructions.

Content filter. We use `gpt-4o-mini-2024-07-18`, the version pinned by Zheng et al. (2025) (Appendix A.1). Zheng et al. do not publish their template; ours mirrors the paper’s wording and asks for a binary decision:

⁵Neuronpedia, <https://www.neuronpedia.org/gemma-2-9b>, source 39-gemmascope-res-16k; accessed May 2026.

Below is a textual description of a feature learned by a sparse autoencoder.
 Decide whether this description is about programming, math, grammar, text
 formatting, or stop words.
 Reply with ONLY 'yes' or 'no'.
 Description: {label}
 Answer:

Features answered “yes” are dropped.

Topic refinement. For each mFETM topic, the top $n + m = 12$ features by $\text{softmax}(\beta_k^*)$ are shown to the model; mVTM uses $\text{softmax}(\beta_k)$. The model flags up to $m = 2$ features that do not belong, and the surviving top $n = 10$ form the topic’s textual representation. We use gpt-5.6-luna. Relative to the F.1 prompt of Zheng et al. (2025), we add context explaining that the labels are automated interpretations of SAE features estimated on general web text, so that a label’s surface domain is not read literally in workplace reviews:

The goal of this task is to evaluate a list of features produced by an automatic method. We call this list a "topic". Given a topic, you’ll be answering the question: "Which feature(s) don’t belong in this list?" For each topic, choose the feature(s) whose meaning does not match with what the list seems to be about.

Reply with a brief reasoning for your choice, and up to two numbers corresponding to the features that don’t belong (or -1 if there are less than two).

Each feature is an automated interpretability description (from Neuronpedia) of a sparse-autoencoder feature, describing the contexts in which it typically activates in general web text. The documents here are employee reviews of companies, where the same feature often activates on workplace analogues of the described context. Judge belonging by what the features would jointly indicate within an employee review, not by the surface domain of any single description. The list is ranked: earlier features have greater topic weight and should count as stronger evidence.

Important: Prioritize identifying feature(s) that are oddly specific and/or clearly out of place. Use your world knowledge in considering whether a feature belongs. If you are not sure, do not choose the feature.

Naming. The feature descriptions alone can be misleading because they summarize where an SAE feature activates in general web text, not how it is used in this corpus. We therefore ground the ten surviving labels with representative reviews. For each topic, we rank documents by the deterministic share $\text{softmax}(\mu_{\theta_d})_k$ and scan the 1,000 highest-share documents in descending order until obtaining 20 unique excerpts containing 5–45 words. The naming model receives only the ranked feature descriptions and excerpt text, never firm, industry, polarity, or document identifiers.

We use the same selection rule for mFETM and mVTM, and use gpt-5.6-luna for naming as well. We then adapt the F.2 prompt of Zheng et al. (2025) as follows:

You name latent topics from a topic model fitted to employee reviews. You receive two ranked evidence sources: automated sparse-autoencoder feature descriptions and representative employee-review excerpts with the highest posterior topic shares. SAE descriptions come from general web text and may be noisy, polysemantic, or expressed through unrelated surface domains. Review excerpts are direct in-corpus evidence, but any single excerpt may contain incidental details.

Infer the narrowest workplace construct supported by repeated patterns across the excerpts and broadly compatible with the features. Prioritize repeated in-corpus meaning when translating a feature description into its workplace analogue. Treat earlier items within each list as stronger evidence.

Rules:

1. Return only a 3--8 word noun phrase, with no preamble or period.
2. Do not name restaurants, shopping, products, customers, family, games, or another surface domain unless that literal subject recurs in the employee-review excerpts.
3. Preserve genuine industry, occupation, policy, or sentiment content only when several high-ranked excerpts support it. Do not name sentiment when it is merely evaluative language around another workplace construct.
4. Do not invent specificity. If the evidence is diffuse, use a transparent broad workplace label rather than joining unrelated fragments.
5. Prefer concrete workplace nouns. Avoid vague filler such as ‘‘elements,’’ ‘‘aspects,’’ ‘‘various,’’ ‘‘references,’’ ‘‘personal recommendations,’’ or ‘‘experiences’’ unless that meaning is explicitly central.
6. Name one central construct.

Names label the output only; estimation never sees them.

E Inference Algorithm

Algorithm 1 states the mFETM estimation procedure. The likelihood uses reparameterized draws of θ , β^* , and the auxiliary coefficients u , which are projected to centered deviations γ . The fixed coefficient-specific precision determines the scale of each u perturbation before centering. The displayed multinomial log likelihood omits the document-specific combinatorial constant. For a minibatch $\mathcal{B}$ from N training documents, we minimize the Monte Carlo per-document negative ELBO

$$-\frac{1}{|\mathcal{B}|} \sum_{d \in \mathcal{B}} \left[\log p(\mathbf{c}_d \mid \tilde{\theta}_d, \tilde{\beta}^*, \tilde{\gamma}) - w \text{KL}_{\theta_d} \right] + \frac{w}{N} (\text{KL}_{\beta^*} + \text{KL}_u),$$

where w is the annealed KL weight. This separates the document-level KL from the two global KL terms. Setting $\gamma \equiv 0$ yields the mVTM likelihood.

Algorithm 1 Mechanistic Fixed-Effects Topic Model (mFETM)

```
1: Input: concept counts  $\{\mathbf{c}_i\}$ , environment indicators  $\{\mathbf{x}_i\}$ ; topics  $K$ , concepts  $V$ , environments  $E$ 
2: Output: topic shares  $\hat{\pi}_d = \text{softmax}(\mu_{\theta_d})$ , global topics  $\hat{\beta}^*$ , centered fixed-effect shifts  $\hat{\gamma}$ 
3: Compute: fit-sample frequencies  $w_e^I, w_e^P$ 
4: Draw once:  $\tau_{e,k,v} \sim \text{Gamma}(a_\tau, b_\tau)$ 
5: Initialize: encoder weights;  $\mu_{\beta^*}, \sigma_{\beta^*}, \mu_u$  with  $\sum_{e \in g} w_e^g \mu_{u_e} = 0$ 
6: for each of 80 training epochs do
7:   sample a minibatch of documents  $d$ 
8:   sample  $\epsilon_\theta, \epsilon_\beta, \epsilon_u \sim \mathcal{N}(0, I)$ 
9:    $(\mu_\theta, \sigma_\theta) \leftarrow \text{encoder}(\mathbf{c}_d)$ 
10:   $\tilde{\theta}_d \leftarrow \mu_\theta + \sigma_\theta \odot \epsilon_\theta$  ▷ draw topic logits
11:   $\tilde{\pi}_d \leftarrow \text{softmax}(\tilde{\theta}_d)$  ▷ topic shares
12:   $\tilde{\beta}^* \leftarrow \mu_{\beta^*} + \sigma_{\beta^*} \odot \epsilon_\beta$  ▷ reparameterize
13:   $\tilde{u}_{e,k,v} \leftarrow \mu_{u_{e,k,v}} + \tau_{e,k,v}^{-1/2} \epsilon_{u_{e,k,v}}$  ▷ auxiliary Gaussian draw
14:   $\tilde{\gamma}_e^g \leftarrow \tilde{u}_e^g - \sum_{e' \in g} w_{e'}^g \tilde{u}_{e'}^g$  ▷ identify each block
15:   $\eta_d \leftarrow \sum_k \tilde{\pi}_{d,k} \text{softmax}(\tilde{\beta}_k^* + \sum_e x_{d,e} \tilde{\gamma}_{e,k})$ 
16:   $\log p(\mathbf{c}_d | \tilde{\theta}, \tilde{\beta}^*, \tilde{\gamma}) \leftarrow \sum_v c_{d,v} \log \eta_{d,v}$ 
17:  compute the Gaussian KL terms for  $\theta$ ,  $\beta^*$ , and  $u$ 
18:  average the likelihood and  $\theta$  KL over the minibatch; divide the global  $\beta^*$  and  $u$  KLs by  $N$ 
19:  clip the total gradient norm at 1; update trainable parameters with Adam; project  $\mu_u$  back to weighted mean zero
20:  if the epoch is a multiple of 5 then
21:    compute validation completion perplexity using the fixed Bernoulli half-split
22:    once  $w = 1$ , retain the parameter state whenever validation perplexity improves
23:  end if
24: end for
25: restore the retained full-KL parameter state
26: return  $\hat{\pi}_d = \text{softmax}(\mu_{\theta_d})$ ,  $\hat{\beta}^*$ , and  $\hat{\gamma}^g = P_g \mu_u^g$ 
```

mFETM is trained for 80 epochs with minibatches of 256 and an exponentially decaying learning rate from 0.01 to 0.001; its KL weight rises linearly from zero to one over the first 40 epochs. mVTM is trained for 60 epochs with the same minibatch size and a learning rate from 0.005 to 0.001; its KL warm-up lasts 30 epochs. Both models use the same validation-based selection rule, with performance assessed every five epochs. We reserve 10% of the original training split for validation and select the full-KL epoch with the lowest document-completion perplexity. We then evaluate the selected checkpoint on the test split without further tuning. Downstream topic shares use $\text{softmax}(\mu_{\theta_i})$; held-out probabilities also use the posterior means of β^* and the centered γ .

F Topic Geometry

We compare the models along two geometric dimensions: separation between topic centroids and dispersion among the leading concepts within a topic. Each SAE feature i has a decoder direction

$\mathbf{W}_{\text{dec}}[i] \in \mathbb{R}^{3584}$ in Gemma’s internal representation. Write $b_k = \beta_k^*$ for mFETM and $b_k = \beta_k$ for mVTM. We construct each topic embedding as the feature-weighted average of these directions:

$$\mathbf{e}_k = \sum_{i=1}^{V_{\text{kept}}} \text{softmax}(b_k)_i \cdot \mathbf{W}_{\text{dec}}[i] \in \mathbb{R}^{3584}.$$

All metrics use this space and the same $V_{\text{kept}} = 4,735$ features, constructed once independently of K . This provides a model-native geometric comparison using the fitted topic weights and SAE decoder space, without external embeddings or additional language-model inference.

F.1 Pairwise Topic Distance

For each model, we compute pairwise cosine distance between topic embeddings:

$$d(k, k') = 1 - \cos(\mathbf{e}_k, \mathbf{e}_{k'}),$$

and average over all $\binom{K}{2} = 1,225$ unordered pairs. A larger mean indicates greater separation among topic centroids.

The mean distance is 0.841 under mFETM (sd 0.127) and 0.889 under mVTM (sd 0.099), a gap of -0.048 . mFETM topic centroids are therefore closer on average (Figure A1). Because pairs share topics, the 1,225 distances are not independent; we therefore report their distribution without treating the pair count as the inferential sample size.

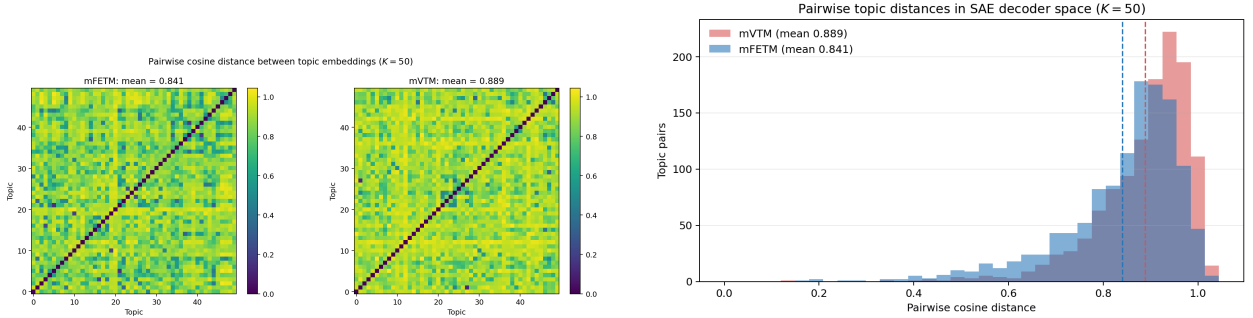


Figure A1: Pairwise cosine distances between topic embeddings at $K = 50$. mFETM topic centroids are closer on average than mVTM topic centroids.

F.2 Within-Topic Semantic Spread

For each topic, we take the 20 concepts with the largest weights under $\text{softmax}(b_k)$. Writing $\mathbf{w}_i$ for concept i ’s decoder direction, topic k ’s spread is the average cosine distance over all pairs of its leading concepts,

$$S_k = \text{mean}_{i < j} [1 - \cos(\mathbf{w}_i, \mathbf{w}_j)],$$

with i and j ranging over 190 pairs. A low S_k indicates a tightly clustered topic; a high value indicates greater geometric dispersion. The distance is bounded between 0 and 2 and lies between 0 and 1 in these data.

Mean spread is 0.9764 under mFETM and 0.9766 under mVTM, a gap of -0.0002 (Welch $p = 0.904$). The within-topic geometry is effectively unchanged across models (Figure A2).

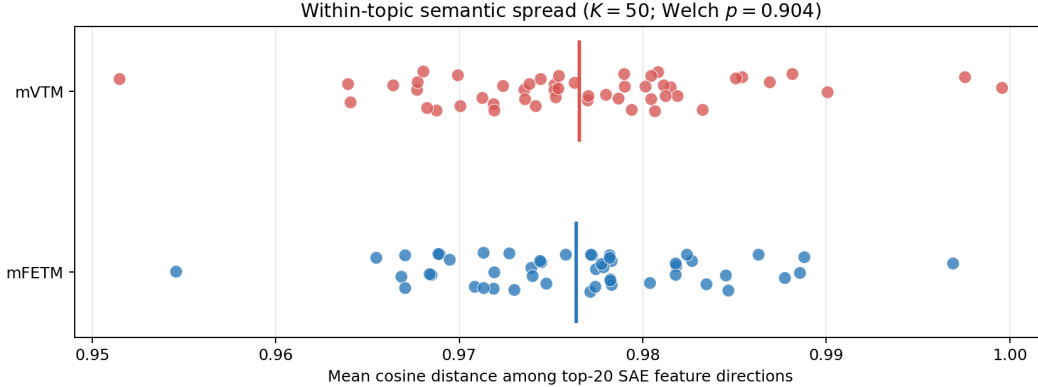


Figure A2: Within-topic semantic spread at $K = 50$. Each dot is one topic; vertical lines mark each model’s mean. The two distributions nearly coincide.

On these measures, mFETM changes the geometry between topics rather than broadening individual topics: within-topic spread is virtually unchanged, while topic centroids are closer together. The centroid shift is consistent with less environment-specific extremity, although greater overlap and redundancy remain observationally equivalent explanations.

G Steering Fluency Check

The faithfulness result is informative only if stronger steering does not simply make the continuation less plausible. We therefore generate a continuation for each of Topics 1–10 at every $\lambda \in \{-400, -300, \dots, 400\}$ using the same prompt and generation procedure as the main exercise. Each continuation is then scored under unsteered Gemma-2-9B. We report the median natural log perplexity across topics with an interquartile band; using the median limits the influence of occasional high-perplexity generations.

Median log perplexity remains between 1.95 and 2.59 across the full path. It equals 2.36 at the direction-specific ablation, falls over most intermediate positive strengths, and reaches 2.59 at $\lambda = 400$. The intervention therefore changes the Pro–Con likelihood gap without producing a general collapse in continuation fit, although the dispersion widens at the strongest positive setting.

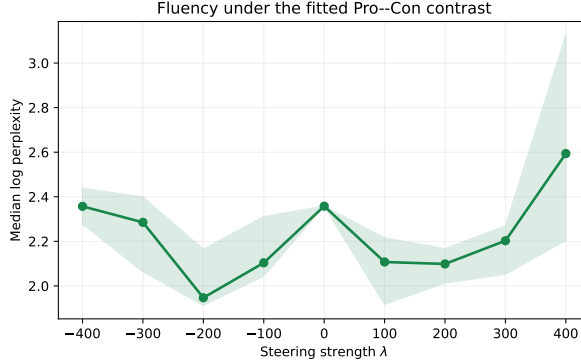


Figure A3: Continuation fluency along the fitted Pro-minus-Con direction. The line reports median log perplexity under unsteered Gemma-2-9B across ten topic-specific generations; the band is the interquartile range. Generation settings match the main steering exercise.

H Removal-Sweep Robustness and Decomposition

The main text reports normalized fit retention because it is directly comparable across models. For the topic-level tables, we estimate one normalization-invariant contrast per topic: the row topic relative to the equally weighted average of the other 49 topics. Re-estimating the logits with the second-largest-share topic omitted instead of the largest-share topic leaves both models’ BH-selected sets unchanged; coefficient and standard-error differences are below 2×10^{-5} . This verifies that the reported topic associations do not depend on the fitting baseline.

With all topics included, mFETM has a pseudo- R^2 of 0.158, compared with 0.136 for mVTM. The firm-clustered, BIC-corrected Vuong test (Vuong, 1989) favors mFETM but does not reject equal relative fit at the 5% level ($z = 1.60$, two-sided $p = 0.109$). In a combined logit, each model’s 49-topic block remains jointly significant conditional on the other (Wald $p = 2.5 \times 10^{-20}$ for mFETM and $p = 2.4 \times 10^{-15}$ for mVTM), indicating that both contain incremental firm-level information.

For the removal exercise, topics are ranked separately within each model by the larger of their absolute pro/con Cohen’s d and standardized cross-industry range. Scores use the full document corpus, with industry ranges restricted to cells containing at least 1,000 documents. The nominal five-point cutoffs remove 2, 5, 8, 10, 12, and 15 of the 50 topics. After each removal, we renormalize the remaining shares and refit the year-fixed-effects logit. At the nominal 5% cutoff, mFETM retains 97.1% of its full-model fit and mVTM 99.5%; from 10% through 30%, mFETM retains substantially more.

Figure A4 reports the companion firm-clustered, BIC-corrected Vuong statistic against a random-removal benchmark. At the nominal 5% cutoff, the targeted statistic is $z = 1.32$ and remains inside the random range. From 10% through 30%, it ranges from 2.68 to 3.38 and exceeds the pointwise 95% random-removal range at every cutoff. From 10% onward, targeted removal shifts relative fit toward mFETM more than removal of the same number of randomly selected topics.

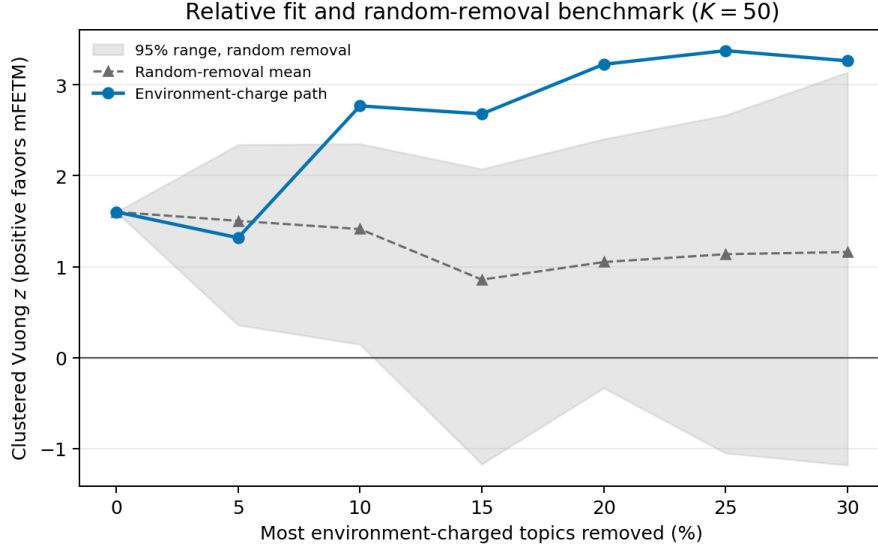


Figure A4: Vuong comparison after removing environment-charged topics, $K = 50$. The targeted line is the firm-clustered, BIC-corrected Vuong z ; positive values favor mFETM. The dashed line and band show the mean and pointwise 2.5th–97.5th percentile range from 20 random-removal draws at each positive cutoff; the 0% point is deterministic. Cutoffs are nominal and use the rounded topic counts reported in the text.

The complementary polarity-only sweep removes one separately ranked topic from each model at every step. The path can be nonmonotone because either removed topic may also carry firm-level signal. Figure A5 pairs the Vuong statistic with both fit paths, allowing movements in relative fit to be traced to losses in mFETM, mVTM, or both. The environment-charge sweep is the primary specification because it evaluates both fixed-effect blocks; the polarity-only sweep isolates the contribution of review-side association.

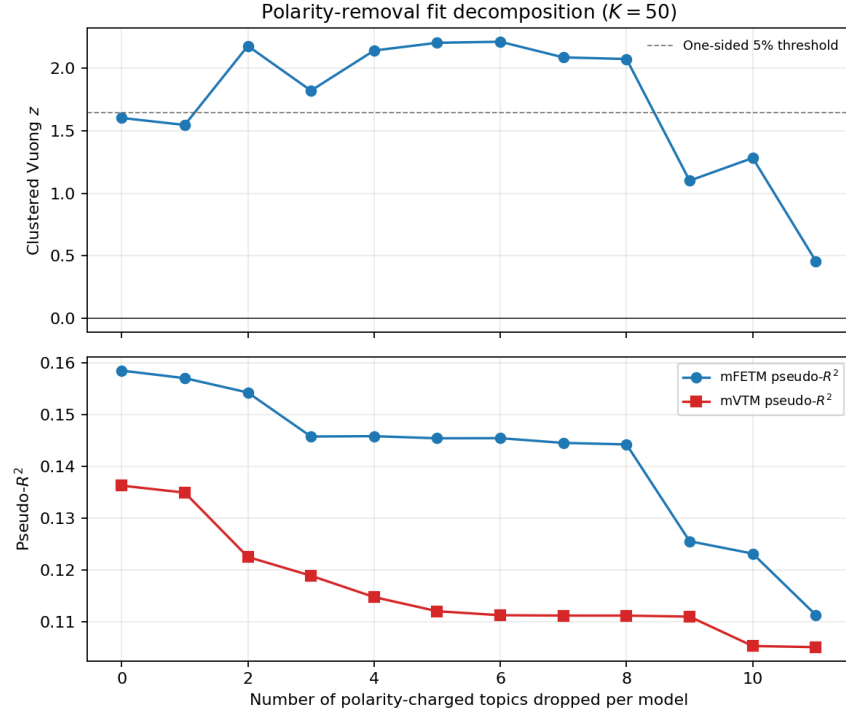


Figure A5: Polarity-removal fit decomposition at $K = 50$. Top: Vuong z as the most polarity-charged topics are removed; the dashed line marks the one-sided 5% threshold, $z = 1.645$. Bottom: each model's pseudo- R^2 over the same sweep. At each step, each model loses its own next most polarity-charged topic, so movements in z are interpreted jointly with both fit paths.